Class 10
CHEMISTRY

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1. ELECTROCHEMISTRY

WORKSHEET – 1

Level – I

1) D	2) A	3) C	4) B	5) D	6) C	7) A
8) B	9) B	10) A	11) C			

3. Cations will discharge at cathode

4. Reduction takes place at cathode

Level – II

1) B	2) D	3) B	4) A	5) D	6) B	7) A
8) D	9) B	10) D	11) B	12) D	13) B	14) D
15) C	16) D					

1.

$Na_2SO_4 \longrightarrow 2Na^+ + SO_4^{2-}$

$2H_2O \rightleftharpoons H_3O^+ + OH^-$

At anode $2H_2O_{(l)} \longrightarrow O_2 + 4H^+ + 4e^-$

At Cathode $2H_2O_{(l)} + 2e^- \longrightarrow H_2 + 2OH^-$

6. $NaCl \longrightarrow Na^+ + Cl^-$

At anode : $2Cl^- \longrightarrow Cl_2 + 2e^-$(Oxidation)

At cathode : $Na^+ + e^- \longrightarrow Na$(Re duction)

7. In case of active electrodes, metal dissolve at anode & deposit at cathode.

8. The ion with lower discharge potential is discharged first (ion with high Reduction potential is discharged first at cathode.)

9. A more active metal than copper will deposit copper from $CuSO_4$.

$Zn_{(s)} + CuSO_{4(aq)} \longrightarrow ZnSO_{4(aq)} + Cu_{(s)}$

10. $CuSO_4 \longrightarrow Cu^{+2} + SO_4^{-2}$

$2H_2O \longrightarrow H_3O^+ + OH^-$

Cu^{+2} has high Reduction potential than H^+so that Cu will discharge at cathode.

OH^-has low Reduction potential than SO_4^{2-}

$2OH^- \longrightarrow H_2O + \frac{1}{2}O_2 + 2e^-$

(or)

$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

$\therefore$ O_2 liberated at anode

So remaining $H^+ + SO_4^{2-} \longrightarrow H_2SO_4$

11. H_2 is liberated at anode when hydrogen is present in the form of H^{-1}

12. In Electrolysis of dilute H_2SO_4, aqueous $AgNO_3$ and Na_2SO_4 using inert electrode (Pt) discharge O_2 at anode because OH^-has high discharge potential than SO_4^{2-} and NO_3^-.

13. After electrolysis of NaCl solution, the resultant solution contain NaOH. So pH increases.

14. Carbon can remove oxygen from XO and YO but not from ZO. Hence " Z " more reactive than X and Y. Y can remove oxygen from XO, So " Y " is more reactive than " X ". So the order of reactivity is $Z > Y > X$.

15. On electrolysis of aqeous $NaCl, H_2$ liberates at cathode because H^+has more reduction potential than Na^+. But on electrolysis of aqueous $CuCl_2, Cu$ deposited at cathode because Cu^+has more reduction potential than H^+

16. Cu is deposited from $CuCl_2$ but from $K_2Cr_2O_7, KMnO_4$ and CH_3COONa solutions H_2 gas is liberated instead of metal.

At anode: $2Cl^- \longrightarrow Cl_2 + 2e^-$(Oxidation)

At Cathode $2H_2O_{(l)} + 2e^- \longrightarrow H_2 + 2OH^-$

At cathode $Cu^{2+} + 2e^+ \longrightarrow Cu$

Level – III

17) A,B,D	18) A,B,C	19) C	20) 4	21) C	22) D
23) A	24) a-p,t; b-q; c-s; d-r		25) A, B	26) B	27) 3
28) A,B,C	29)a-q,s; b-p,q; c-p,r; d-p,q		30) C		

19. Degree of dissociation

$$\propto \frac{1}{\text{Concentration of solution}}$$

22. NH_2CONH_2 (Urea) covalent compound. It doesn't produce ions.

23. All Organic acids are weak electrolytes

WORKSHEET – 2

Level – I

1) C	2) C	3) C	4) C	5) B	6) A	7) C
8) A	9) C					

Level – II

1) A	2) B	3) B	4) B	5) D	6) C	7) A
8) B	9) C	10) D	11) D	12) C	13) B	14) A
15) B	16) C	17) B	18) B	19) B		

1. $Ag^{+1} + le^{-} \longrightarrow Ag$

   $1\ F \longrightarrow 108\ g$

2. $Al^{+3} + 3e^{-} \longrightarrow Al$

   $3\ F \longrightarrow 27\ g$

   $1.5\ F \longrightarrow ?$

   $= \frac{1.5 \times 27}{3} = 13.5\ g$

3. The ion which carries one unit charge will be deposited by 1 F of charge

4. $Mg^{+2} + 2e^{-} \longrightarrow Mg$

   $2 \times 6.023 \times 10^{23}$

   e^{-s} ___ Imole of Mg

5. $e = \frac{E}{96{,}500}$

6. $Mg^{+2} + 2e^{-} \longrightarrow Mg$

   $2Ne^{-1}$ s_24 g

   ? 3 g

   $= \frac{3 \times 2\ N}{24} = \frac{N}{4}$ electrons

7. $\frac{w_1}{w_2} = \frac{E_1}{E_2}$

8. $e = \frac{E}{96500}$

9. $\frac{w_1}{w_2} = \frac{E_1}{E_2}$

10. $Cu^{+2} + 2e^{-} \longrightarrow Cu$

    $= \frac{4 \times 63.5}{2}$

    $= 127\ g$

    $2\ F$

    $63.5\ g$

    $4\ F \longrightarrow ?$

11. $\frac{w_1}{w_2} = \frac{E_1}{E_2}$

12. $Ag^{+} + e^{-} \longrightarrow Ag$

    $Cu^{+2} + 2e^{-} \longrightarrow Cu\ 2H^{+} + 2e^{-} \longrightarrow H_2$

    $IF \longrightarrow 1$ mole

    $2\ F$

    1 mole

    $2\ F \longrightarrow 1$ mole

    $1\ F_{-\frac{1}{2}}$ mole

IF $\longrightarrow \frac{1}{2}$ mole

$n_{Cu}: n_{Ag}: n_{H_2} = \frac{1}{2}: 1: \frac{1}{2} = 1: 2: 1$

13. $Ag^+ + e^- \longrightarrow Ag$

6.02×10^{23} electrons $\longrightarrow$ 108 g

$? \longrightarrow 10.8$ g

$= \frac{6.02\times10^{23}\times10.8}{108} = 6.02 \times 10^{22}$ electrons

14. $w = \frac{\text{A.wt} \times i \times t}{\text{valency} \times 96500}$

15. $Q = i \times t$

$= 100 \times 10^{-3} \times 30 \times 60 = 180C$

16. $Cr_2^{(+6)}O_7^{-2} \longrightarrow 2Cr^{+3}$

Change in oxidation number = 3

No. of electrons involved $= 2 \times 3 = 6$

$\therefore$ charge required = 6 F

17. $w = \frac{\text{A.wt}\times i\times t}{\text{valency} \times 96500} = \frac{63.5\times4\times2\times60\times60}{2\times96500} = 9.47$ g

Current efficiency

100?

Cu deposited $9.47 - 6$

$= \frac{6}{9.47} \times 100$

$= 63.3\%$

18. $V = \frac{V_e\times i\times t}{96500}$

$67.2 = \frac{11.2 \times i \times 15 \times 60}{96500}$

$\therefore$ i = 643.33amp

19. $V = 20 \times 20 \times 10^{-2} = 4\ cm^3$

$d = \frac{m}{v} \Rightarrow m = d \times v = 8.94 \times 4 = 35.76$ g of Cu

$Cu^{+2} + 2e^- \longrightarrow Cu$

63.5 g of Cu_2 $\times$ 96500

35.76 g of Cu ?

$= \frac{35.76 \times 2 \times 96500}{63.5} = 108687C$

Level – III

20) A,B,D	21) A,B,D	22) B	23) 2	24) a-r; b-s; c-p; d-q	
25) C	26) B	27) C	28) A	29) 4	30) B,C

20. $Ca^{+2} + 2e^- \longrightarrow Ca$ 2 F_40 g of Ca.

21. $Ag^+ + e^- \longrightarrow Ag$ $Cu^{+2} + 2e^- \longrightarrow Cu$ $Al^{+3} + 3e^- \longrightarrow Al$ IF $\longrightarrow$ 1 mole 2 F $\longrightarrow$ 1mole 3 F $\longrightarrow$ 1 mole IF $\longrightarrow$? $1F$? $= \frac{1}{2}$ mole $= \frac{1}{3}$ mole

$n_{Ag} : n_{Cu} : n_{Al} = 1 : \frac{1}{2} : \frac{1}{3}$

26. 1 F_1 eq of an element

96,500_1 eq of an element

?__ 0.1 eq of an element

$= \frac{96500 \times 0.1}{1} = 9650C$

27. $\frac{w_1}{w_2} = \frac{E_1}{E_2}$

28. 1C of current will deposit electrochemical equivalent of an element $w = e \times i \times t$

$w = w \times i \times 0.25$

$1 = i \times 0.25$

$i = 4amp$

30. $H_2O \longrightarrow H_2 + \frac{1}{2}O_2$

18 g $\longrightarrow$ 2 F $2H^+ + 2e^- \longrightarrow H_2$

90 g $\longrightarrow$?

(2 F)

= 10 F

Since current is of 50% efficiency, the electricity required = 20 F

$H_2O \longrightarrow H_2 + \frac{1}{2}O_2$

18 g $\longrightarrow 22.4 + 11.2 = 33.6$ L

90 g $\longrightarrow$?

$= \frac{90 \times 33.6}{18} = 168$ L of gases.

WORKSHEET – 3

Level – I

1) D	2) A	3) B	4) A	5) C	6) C	7) C
8) B	9) A	10) C	11) A			

11. $R = e \times \frac{l}{a}$

$\frac{1}{C} = \frac{1}{K} \times \frac{l}{a}$

Given that $C = K$ Hence $\frac{1}{a} = 1$

Level – II

1) A	2) B	3) A	4) C	5) A	6) B	7) B
8) C	9) A	10) C	11) B	12) B	13) B	14) B
15) A	16) B	17) D	18) B	19) A	20) A	

3. $\Lambda = \frac{k \times 1000}{N} = \frac{1000}{e \times N} = \frac{1000}{Rc}$

4. $\Lambda \propto \frac{1}{N}$

5. $\Lambda = \frac{K \propto 1000}{1} = 10^3 \times K$

6. $\Lambda_m = \frac{K \times 1000}{m} = \frac{1.061 \times 10^{-4} \times 10^3}{10^{-2}} = 1.061 \times 10 = 10.61$

7. $R = e \times \frac{l}{a}$

$$R = \frac{1}{K} \times \frac{l}{a}$$
$$K = \frac{1}{R} \times \frac{l}{a}$$
$$= \frac{1}{2.5 \times 10^3} \times 1.25$$
$$= 0.5 \times 10^{-3}$$
$$= 5 \times 10^{-4}$$
$$\Lambda = \frac{K \times 1000}{N} = \frac{5 \times 10^{-4} \times 10^3}{10^{-1}} = 5$$

12. $R = \frac{1}{K} \times \frac{l}{a}$

$$82.4 = \frac{1}{2.768 \times 10^{-3}} \times \frac{1}{a}$$

$$\frac{l}{a} = 82.4 \times 2.768 \times 10^{-3} = 0.2280 \text{ cm}^{-1}$$

13. $R = e \times \frac{l}{a}$

$$50 = \frac{1}{K} \times \frac{0.02}{0.0004}$$

$$K = \frac{1}{50} \times \frac{0.02}{0.0004} = 1 \text{Sm}^{-1}$$

15. $C = \frac{1}{R} = \frac{1}{210} = 0.00476 = 4.76 \times 10^{-3} \text{mho}$

16. The size of hydrated cations is $Li^+_{(aq)} > Na^+_{(aq)} > K^+_{(aq)}$

So order of equivalent conductance is KCl > NaCl > LiCl

17. $\Lambda_m = \frac{K \times 1000}{m} \Rightarrow K \propto m$

18. $\alpha = \frac{\Lambda^c}{\Lambda^\infty} = \frac{10}{200} = \frac{1}{20}$

$$\alpha\% = \frac{1}{20} \times 100 = 5\%$$

20. $\Lambda_{BaSO_4} = \Lambda_{BaCl_2} + \Lambda_{H_2SO_4} - \Lambda_{HCl} = X_1 + X_2 - X_3$

Level – III

21) A,B,C,D 22) A,B,C 23) A 24) C 25) B

26) i-p,s; ii-r; iii-t; iv-q 27) 6 28) A 29) B,C

30) A,B,C

21. $G^* = \frac{l}{a} = \frac{0.5}{2} = 0.25\ cm^{-1}$

22. $\Lambda_m = \frac{K \times 1000}{m}$

$200 = \frac{K \times 1000}{0.04} \Rightarrow K = \frac{200 \times 4 \times 10^{-2}}{10^3} = 8 \times 10^{-3}$

$K = C \times G^*$

$8 \times 10^{-3} = C \times 0.25$

$C = \frac{8 \times 10^{-3}}{0.25} = 32 \times 10^{-3} = 0.032$

25. $C = \frac{1}{R} \Rightarrow R = \frac{1}{C} = \frac{1}{0.032} = 31.25\Omega$

$I = \frac{V}{R} = \frac{5}{31.25} = 0.16\ A$

27. $\Lambda_m^0 = \Lambda_{eq}^0 \times$ charge of ion $= \Lambda_{eq}^0 \times 6$

$\therefore x = 6$

30. $\mu_{K^+} = \frac{\lambda_{K^+}}{96500} = \frac{96.5}{96500} = 10^{-3}$

Speed of ion $= \frac{\text{Ionic Mobility}}{\text{Potential gradient}}$ Potential gradient $= \frac{\text{P.D. between electrodes}}{\text{Distance between electrodees}}$

$= \frac{10^{-3}}{1} = 10^{-3} = \frac{10}{10} = 1$

$1sec \longrightarrow 10^{-3}\ cm \cdot sec^{-1}$

$5 \times 10^3 sec \longrightarrow ?$

$= 5 \times 10^3 \times 10^{-3} = 5$ cm.

WORKSHEET – 4

Level – I

1) A 2) B 3) B 4) C 5) A 6) B 7) A

8) C 9) B 10) D

Level – II

1) B 2) B 3) D 4) A 5) A 6) C 7) C

8) A 9) C 10) C 11) B 12) C 13) B 14) B

15) B

Level – III

16) A, B	17) A, B	18) A	19) A	20) B	21) C
22) a-4; b-3,5; c-1; d-2		23) 0	24) 1,2,3,4	25) A, B	26) C
27) A	28) A	29) A	30) B		

WORKSHEET – 5

Level – I

1) C	2) B	3) A	4) C	5) C	6) D	7) B
8) B	9) B	10) D				

Level – II

1) C	2) D	3) C	4) D	5) D	6) A	7) B
8) C	9) A	10) A	11) A	12) A	13) D	14) A
15) A	16) A	17) C	18) D	19) A	20) A	21) C
22) C	23) A	24) A				

13. $2Cu^{+} \longrightarrow Cu^{2+} + Cu$

Cell I:	Cell II
$Cu^{+} \longrightarrow Cu^{2+} + e^{-}$	$Cu \longrightarrow Cu^{2+} + 2e^{-}$
$Cu^{+} + e^{-} \longrightarrow Cu$	$2Cu + 2e^{-} \longrightarrow 2Cu$
$2Cu^{+} \longrightarrow Cu^{2+} + Cu$	$2Cu^{+} \longrightarrow Cu^{2+} + Cu$
$E^{0} = -0.15 + 0.50$	$E^{0} = -0.325 + 0.50$
$= +0.35$ V	$= +0.175$v
n = 1	n = 2

15. Zinc electrode acts an anode

Hydrogen electrode acts as a cathode

Because zinc reduction potential value less it mean ' Zn ' is more reactive than ' H '

Anode : $2OH^{-} \rightarrow H_2O + \frac{1}{2}O_2 + 2e^{-}$

Cathode: $2H^{+} + 2e^{-} \rightarrow H_2$

16. $Fe^{3+} + e^{-} \longrightarrow Fe^{+2}; E^{0} = 0.771$ V

$I_2 + 2e^{-} \longrightarrow 2I^{-}; E^{0} = 0.536$ V

E^{0} cell for the cell reaction

here two reactions are reduction reaction

$E_{cell} = E_R - E_L$

$= 0.771 - 0.536$

$= 0.235$ V

17. $Pt, H_2/HA_2//HA_2/H_2$ (1atm)

$H_2/H^+_{c_1}//H^+_{c_2}/H_2$

$E_{cell} = -0.059\left(P_{K_{\sigma_1}} - P_{K_{\alpha_2}}\right)$ $E_{cell} = -0.059(3-5)$

$= 0.118$

19. $E^0_{RppyZ^-} > E^0_{Rpyy-} > E^0_{Rpyx-x}$

Thus, order of oxidising power will be $Z > Y > X$

20. E^0_{op} for Zn = +0.762 V (maximum in given values)

more positive is E^0_{OP}, more is the tendency to g at itself oxidised or strong reducing agent.

21. $Fe^{+3} + I^{-1} \rightarrow Fe^{+2} + \frac{1}{2}I_2$

$I_2/I^-//Fe^{+3}/Fe^{+2}$

$E^0_{cell} = E^0_{cathode} - E^0_{anode} = E^0_{Right} - E^0_{Left}$

$E^0_{cell} = 0.77\text{ V} - 0.34\text{ V}$

$= 0.23$ V

22. Emf of cell = 1.1 V

Reduction potential E^0_R of $Zn^{2+}/Zn = -0.78$ V

Oxidation potential of Cu/Cu^{2+} = ?

$E^0_{cell} = E_R - E_L$

$E_{1.1} = E_{Red} - (-0.78)$

$= 0.32$ V

$Cu/Cu^{2+} E^0_{0\times1} = -0.32$ V

23. E^0_{RP} for Ag/Hg and Cu are $-0.80, -0.79, -0.34$. Mg^{2+} is not discharged in aqueous solution.

24. $[2e + Fe^{2+} \longrightarrow Fe] \times 3$

$[(Fe^{3+})_2 + 6e \longrightarrow 2Fe]1$

Thus, ratio of $Fe^{3+}: Fe^{2+}$ is 2: 3.

Level – III

25) A,C,D 26) A 27) A 28) D 29) 0.48

30) a-q; b-p,r; c-p,q; d-s 31) 1.298 32) A, B, D 33) a-s; b-r; c-p; d-q

WORKSHEET – 6

Level – I

1) B 2) A 3) C 4) D 5) B 6) A 7) B

8) A 9) B 10) A

10. $E_{RP} = E^0_{RP} + \frac{0.059}{n}\log\frac{[M^{x+n}]}{[M^{x+}]}$

Level – II

1) A	2) A	3) C	4) A	5) B	6) D	7) A
8) C	9) C	10) D	11) A	12) A	13) D	14) B
15) A	16) B					

7. Anode : $H_2 + 2H_2O \longrightarrow 2H_3O^+ + 2e$

cathode: $Cl_2 + 2e^- \longrightarrow 2Cl^-$

$E^0_{cell} = \frac{0.059}{n}\log k = \frac{0.059}{2}\log_{10} 10^{46}$

$= 1.357v$

8. $E_{RP} = E^0_{RP} + \frac{0.059}{4}\log\frac{[TeO_3^{2-}]}{[OH^-]^6}$

$[Te^{4+} + 4e^- \rightarrow Te$; Also pH = 12 $\therefore [OH^-] = 10^{-2}]$

$\therefore E_{RP} = -0.57 + \frac{0.059}{4}\log\frac{1}{(10^{-2})^6} = -0.393$ V

9. Anode $H_2 \longrightarrow 2H^+ + 2e^-$

$[H^+] = 10^{-6}M$

Cathode $2H^+ + 2e^- \longrightarrow H_2O$

$[H^+] = aM$

$E_{cell} = E_{op/H} + E_{RPH^+/H}$

$= E^0 - \frac{0.059}{2}\frac{\log[H^+]^2 \text{ cathode}}{[H^+]^2 \text{ anode}}$

$-\frac{0.059}{2}\log\frac{[H^+]\text{cathode}}{10^{-6}}$

$[H^+]$cathode = 10^{-4}M.

10. $\Delta G^0 = -nFE^0$

from given data

i) $Cu_{(s)} \longrightarrow Cu^{2+}_{(aq)} + 2e^-$

$\Delta G^0_1 = -2(-0.34) \times F$

ii) $Cu^{2+}_{(aq)} + e^- \longrightarrow Cu^{+}_{(aq)}$

$\Delta G^0_2 = -1(0.15) \times F$

$\Delta G^0_3 = \Delta G^0_1 + \Delta G^0_2$

$-n_3FE^0 = -n_1FE^0_1 - n_2FE_2$

$-E^0 = -2(-0.34) - 1(0.15)$

$= (-2 \times -0.34) + (-1 \times 0.15)$

$-E^0 = +0.68 - 0.15 = 0.531 E^0 = -0.53v$

Reaction $2Cu^{+}_{(aq)} \rightleftharpoons Cu^{2+}_{(aq)} + Cu_{(s)} E^0 = ?$

$Cu^{+}_{(aq)} + e^{+} \rightleftharpoons Cu_{(s)}, E^0 = 0.53v$

$Cu^{+}_{(aq)} \rightleftharpoons Cu^{2+}_{(aq)} + e^{-}$

$E^0 = 0.15v$

$2Cu^{+}_{(aq)} \rightleftharpoons Cu^{2+}_{(aq)} + Cu_{(s)}$

$E^0 = +0.38v$

11. $E = \frac{0.059}{3} \log \frac{[Cr^{3+}] \text{ R.H.S}}{[Cr^{3+}] \text{ L.H.S}}$

$= \frac{0.059}{3} \log \frac{1}{[0.04]} = 0.028 \text{ V}$

12. $E^0_{Zn^{+2}/Zny^{2-}/y^{4-}} = E^0_{Zn^{2+}/Zn} + \frac{0.059}{2} \log k$

$K_f = \frac{[Zny^{2-}]}{[y^{4-}]} \therefore K = \frac{1}{Kf}$

$\therefore E^0_{Zn^{2+}/Zny y^{2-}/y^{4-}} = -0.76 + \frac{0.059}{2} \log \frac{1}{3.2 \times 10^{16}}$

$= -1.25 \text{ V}$

16. If the cathode concentration is more e.m.f of the cell less. If anode concentration is more, e.m.f of the cell is more.

17. Concentration of $[H^+]$in anode -10^{-8}M concentration of $[H^+]$in cathode- 0.001/

At Anode $= \frac{1}{2} H_2 \longrightarrow H^+ + e^-$

$$10^{-3} \; At \qquad Cathode = H^+ + e^- \longrightarrow \frac{1}{2} H^2$$
$$H^+ \longrightarrow H^+$$

Hence $n = 1$

$E_{cell} = E^0_{cell} - \frac{0.06}{1} \log \frac{10^{-8}}{10^{-3}}$

$= 0 - 0.0610 \text{ g} 10^{-5}$

$= 0 + 0.3010g10$

$= 0.300$

$\simeq 0.295$

Level – III

18) D 19) B,C,D 20) A 21) C 22) C

23) a-r; b-p; c-v; d-t; e-s; f-q; g-u 24) 0.8362 25) A 26) D

27) B 28) C 29) D 30) C

22.

$$Cr \longrightarrow Cr^{3+} + 3e^-$$
$$Fe^{2+} + 2e^- \longrightarrow Fe$$
$$2Cr + 3Fe^{2+} \longrightarrow 2Cr^{3+} + 3Fe$$
$$E^0_{RP} = -0.72v$$
$$E^0_{RP} = -0.42v$$

According to Nernst equation

$$E_{cell} = E^0_{cell} - \frac{0.059}{n}\log_{10}\frac{[Cr^{3+}]^2}{[Fe^{2+}]^3}$$
$$= 0.3 - \frac{0.59}{6}\log_{10}\frac{(0.1)^2}{(0.01)^3}$$
$$= 0.3 - \frac{0.059}{6}\log_{10}10^4$$
$$= 0.3 - 0.039$$
$$E_{cell} = 0.261v$$

WORKSHEET - 07

Level – I

1) A 2) B 3) A 4) A

Level – II

1) D	2) D	3) B	4) A	5) B	6) B	7) A
8) B	9) A	10) B	11) A	12) A	13) A	14) A
15) A	16) A	17) C	18) A			

4. $Cr^{3+} + 3e^- \longrightarrow Cr;\ -\Delta G^0_1 = 3 \times -0.74 \times F$

$Cr^{3+} + e^- \longrightarrow Cr^{2+};\ -\Delta G^0_2 = 1 \times -0.40 \times F$

$Cr^{2+} + 2e^- \longrightarrow Cr;\ -\Delta G^0_3 = 2 \times E^0 \times F$

$= (3 \times -0.74 + 1 \times 0.40)F = -1.82\ F$

$\therefore E^0 = -0.91\ V$

5. $E^{0-}_{Mn^{3+}/Mn^{2+}} = 1.57\ V$

$E^0_{Fe^{3+}/Fe^{2+}} = 0.77\ V$

$E^0_{CO^{3+}/CO^{2+}} = 1.97\ V$

$E^0_{Cr^{3+}/Cr^{2+}} = -0.41\ V$

7. $\Delta G = -nFE^0_{\text{cell}}$

$= -2 \times 96{,}500 \times 0.82v$

$= -158{,}260j$

$= -158.26KJ$

$K_c = 10^{\left(\frac{n \times E^0_{\text{cell}}}{0.059}\right)}$

$K_c = 10^{\left(\frac{2 \times 0.82}{0.059}\right)}$

$K_c = 6.2 \times 10^{27}$

8. $-\Delta G^0 = nE^0\ F\ (\because n = 2)$

$E^0 = \dfrac{65.62 \times 10^3}{2 \times 96500} = 0.34V$

9. $E^0_{\text{Cell}} = E^0_{\text{OPFe}} + E^0_{\text{RPH}_2\text{O}} = 0.44 + 1.23 = 1.67\ \text{V}$

$\therefore \Delta G^0 = -nE^0\ F = -2 \times 1.67 \times 96500\ \text{J}$

$= -322.31\text{KJmol}^{-1}$

11. $K_c = 10^{\left(\frac{n \times E^0_{\text{cell}}}{0.059}\right)}$

$K_c = 10^{\left(\frac{2 \times 0.46}{0.059}\right)}$

$K_c = 3.69 \times 10^{15}$

12. $Cu^{+2} + 2e^- \longrightarrow Cu; \Delta G^0_1 = (-2)(F)(0.339) \longrightarrow (1)$

$Cu^{+2} + e^- \longrightarrow Cu^+; \Delta G^0_2 = -(1)(F)(0.153)$

$Cu^+ \longrightarrow Cu^{+2} + e^-; \Delta G^0_2 = (F)(0.153) \longrightarrow$

$\text{Eq}(1) + (2)$

$Cu^+ + e^- \longrightarrow Cu; \Delta G^0_3 = -(F)\left(E^0_{\text{cell}}\right)$

$-0.678F + 0.153F = -FE^0_{\text{cell}}$

$E^0_{\text{cell}} = +0.525\ \text{V}$

13. $In^{+2} + e^- \longrightarrow In^+, \Delta G^0 = -1 \times F(0.4)$

$In^+ \longrightarrow In^{+3} + 2e^-, \Delta G^0 = -2\ F(-0.42)$

$In^{+2} \longrightarrow In^{+3} + e^-(\text{oxidation})$

$-1 \times FE^0 = 0.44$ (oxidation potential)

RP of anode is -0.44 V

Now $E^0_{\text{cell}} = E^0_c - E^0_A$

$= 0.15 - (-0.44)$

$= 0.15 + 0.44$

$= 0.59$

At equilibrium, $0.59 = \frac{0.059}{1} \log k$

$k = 10^{10}$

14. $E^0_{\text{Cell}} = E^0_{\text{OPFe}} - E^0_{\text{OPCu}}$

$= 0.44 - (-0.337)$

$= 0.777\ \text{V}$

$E_{\text{cell}} = E^0_{\text{cell}} - \frac{0.059}{2}\log k$

At equilibrium $E_{\text{cell}} = 0$

$= 0.777\ \text{V} = \frac{0.059}{2}\log K_c$

$K_C = 2.18 \times 10^{26}$

15. $\Delta G^0 = -nFE^0_{\text{cell}}$

$E^0_{\text{cell}} = 0.34 - (0.76)$

$= 1.10\ \text{V}$

$\Delta G^0 = -2 \times 1.1 \times 96487$

$= -212.3\text{KJ}$

16. $E^0_{Fe^{3+},Fe^{2+}/Pt} = 0.771\ \text{V}$

$E^0_{Sn^{4+},Sn^{2+}/Pt} = 0.150\ \text{V}$

$E^0_{\text{cell}} = 0.771 - 0.15$

$= 0.62\ \text{V}$

$E^0_{\text{cell}} = \frac{0.059}{2}\log kc$

$0.62 \times 2 = 0.059\log kc$

$\text{Log}k_c = 21.0169$

17. $E^0_{Ni^{2+}/Ni} = -0.25\ \text{V}$

$E^0_{Ag^+/Ag} = 0.8\ \text{V}$

$E^0_{\text{cell}} = 0.8 - (-0.25)$

$= 1.05\ \text{V}$

$E^0_{\text{cell}} = \frac{0.059}{2}\log K_c$

$1.05 = \frac{0.059}{2}\log K_c$

$\log K_c = \frac{1.05 \times 2}{0.059}$

$\log K_c = 35.59$

$K_c = 10^{35.59}$

$K_c = 3.3 \times 10^{35}$

18. $E^0_{Zn^{2+}/Zn} = -0.76\ \text{V}$

$E^0_{Sn^{2+}/Sn} = -0.14\ \text{V}$

$F = 96500\text{cmol}^{-1}$

$E^0_{\text{cell}} = E_R - E_L$

$= -0.14 - (-0.76)$

$= 0.62\ \text{V}$

$\Delta G^0 = W = -nFE^0$

$= -2 \times 96500 \times 0.62$

$= -119.7\text{KJ}$

Level – III

19) A	20) A,C,D	21) B,C	22) B	23) C
24) a-p,t; b-r; c-q,s; d-p,t		25) C	26) D	27) B,C
28) A, B, C	29) B	30) B		

19. $-\Delta G^0 = nE^0$ F

$\therefore E^0 = \frac{-\Delta G^0}{nF}$ since, ΔG^0 is

Extensive property and then E^0 is also intensive property

29. E^0_{cell} of $Zn/Zn^{2+} \parallel Cu^{2+}/Cu = 1.1$ V

$E^0_{cell} = \frac{0.059}{2} \log kc$

$\log K_c = \frac{1.1 \times 2}{0.059}$

$\log K_c = \frac{2.2}{0.059}$

$K_c = 1.9 \times 10^{37}$

The order is = 10^{37}

Here, n = 4 from the equation

2. METALLURGY

WORKSHEET 01

Level – I

1) C	2) C	3) B	4) A	5) B	6) B	7) C
8) A	9) D	10) D	11) A			

1. The isolation and extraction of metals from their ores
2. The minerals from which metal can be economically and conveniently extracted are called ores.
3. Rocky impurities present in the mineral are called matte
4. Bauxite $Al_2O_3 \cdot 2H_2O$ (Major ore of Al)
5. Pyrolusite $-MnO_2$
6. Galena - PbS
7. Rock salt (NaCl) is the ore of Sodium
8. Gypsum-Ca, Dolomite-Mg, Magnesite-Mg, Carnalite-Mg
10. Haematite Fe_2O_3,Zincite = ZnO, Iron pyrites FeS_2, Galena PbS Zinc Blende ZnS, Siderite $FeCO_3$, Calamine

Level – II

1) A	2) A	3) B	4) B	5) A	6) A	7) A
8) B	9) D	10) 2	11) 7	12) C	13) D	14) A
15) D	16) 6	17) B	18) A	19) C	20) A	21) C
22) C						

1. An ore is usually contaminated with earthy or undesired materials known as gangue.
2. magnesite $-MgCO_3$, bauxite-oxide ore, Gypsum - sulphate ore , Galena-sulphide ore
3. Gypsum- $CaSO_4 2H_2O$
6. Pyrolusite $-MnO_2$ (Oxide ore)
9. Zinc blende, ZnS (ore of zinc) etc., and is based on the fact that gangue and ore particles have different degree of wettability with water and pine oil; the gangue particles are preferentially wetted by water while the ore particles are wetted by oil. In this process one or more chemical frothing agents are added.
10. Bauxite $-Al_2O_3 \cdot 2H_2O$
11. Epsom salt $-MgSO_4 \cdot 7H_2O$
12. Amlagam $-Ag_2Hg$
13. The magnetic impurity present in cassiterite (SnO_2) ore is Wolframite- $(Fe, Mn)WO_4$
14. Copper pyrites ore is sulphide ore therefore,

Froth Floatation method is mainly employed for the concentration of sulphide ores.

15. "Reduction of the oxide with carbon at high temperature is known as smelting". The oxides of less electropositive metals like Pb, Zn, Fe, Sn, Cu etc. are reduced by strongly heating them with coal or coke, in the blast furnace.
16. $MgCl_2 \cdot KCl \cdot 6H_2O$
17. The magnetic oxide of iron is Magnetite - Fe_3O_4
20. Carnalite is $KCl \cdot MgCl_2 \cdot 6H_2O$

Level – III

23) A 24) A 25) D 26) C 27) a-s; b-r; c-p; d-q
28) A,B,C 29) a-s; b-s; c-p,q; d-r 30) A,B,D

27. a) Sederite –Fe
    b) Sphalerite-Zn
    c) Cryolite-Al
    d) Dolomite - Ca

WORKSHEET 02

Level – I

1) B 2) D 3) D 4) A 5) C 6) A 7) D
8) D 9) D 10) B 11) B 12) A 13) B 14) B

1. The removal of the excess sulphur contained in sulphide ores in the form of SO_2 then in the by heating in an excess of air is called roasting.
2. "Reduction of the oxide with carbon at high temperature is known as smelting". The oxides of less electropositive metals like Pb, Zn, Fe, Sn, Cu etc. are reduced by strongly heating them with coal or coke, in the blast furnace.
   Metal Oxide + Coke + flux $\longrightarrow$ Metal $+CO_2 \uparrow$ + Slag
4. Metals of very high purity are obtained by zone refining.
   This refining method is based on the fact that impurities tend to remain dissolved in molten metal. Ge, Si and Ga used as semiconductors are refined in this manner.
5. Polling-purification of Cu, Cupellation-purification of Ag,
   Van Arkel method-purification of Ti, Zr, Mond's process - purification of Ni
6. Van Arkel method-purification of Ti, Zr
7. Mond's process - purification of Ni
8. $Cr_2O_3 + Al \longrightarrow Al_2O_3 + 2Cr; Fe_2O_3 + Al \longrightarrow Al_2O_3 + 2Fe$
9. Electrolytic reduction method is mainly used for the extraction of highly electropositive metals. IA, IIA & AI
12. Bell metal –80% Copper & 20% tin

Level – II

1) A	2) D	3) B	4) D	5) A	6) B	7) C
8) A	9) A	10) 3	11) B	12) D	13) B	14) C
15) 2	16) B	17) B	18) A	19) C	20) A	21) A
22) C	23) A					

1. Calcination is a process in which ore is heated, generally in the absence of air, to expel water from a hydrated or hydroxide ore and oxide or carbon dioxide from a carbonate ore at temperature below their melting points.
2. The removal of the excess sulphur contained in sulphide ores in the form of SO_2 then by heating in an excess of air is called roasting.
3. Cupellation : This process is used to purify silver containing the impurities of Pb.This process is used when impurity have greater affinity towards O_2 while metal does not have.
5. Distillation : Metals having low boiling point are refined by this method, for example, zinc, cadmium and mercury.
6. Zone refining : Metals of very high purity are obtained by zone refining. Ge, Si and Ga used as semiconductors are refined in this manner.
11. The main function of roasting is Oxidation .
12. Liquation method is used for the refining of metals having low melting point and are associated with high melting impurities.
    Examples, Pb, Sn, Sb and Bi.
16. The removal of the excess sulphur contained in sulphide ores in the form of SO_2 then by heating in an excess of air is called roasting.
17. Employed for purification of metals like titanium and zirconium.
    $Ti(s) + 2I_2(g) \overset{523k}{\longrightarrow} TiI_4(g)$
    Impure
    $TiI_4(g) \overset{1700k}{\longrightarrow} Ti(s) + 2I_2(g)$
18. Rolled gold is an alloy of copper & aluminium
19. Calcination is a process in which ore is heated, generally in the absence of air, to expel water from a hydrated, or hydroxide ore and oxide or carbon dioxide from a carbonate ore at temperature below their melting points.
21. Refractory metals are used in construction of furnaces because they can withstand high temperature
22. The part of furnace where we kept fuel for burning is firebox

Level – III

24) B,C 25) A 26) D 27) C 28) a-s; b-q; c-r; d-s
29) A,C 30) a-s; b-q; c-r; d-p

18. 24.Electrolytic reduction is a process mainly used for the extraction of highly electropositive metals. I A, II A& Al

28. Liquation is a method is used for the refining of metals having low melting point and are associated with high melting impurities.
Distillation: It is a process metals having low boiling point are refined by this method, for example, zinc, cadmium and mercury.
Cupellation : This process is used to purify silver containing the impurities of Pb.